Database Normalization

Normalization is the process of organizing data to reduce redundancy and improve integrity.

In a well-designed database, each fact should appear in one place. This makes updates easier and re

1NF: Each table cell contains a single value and each row is unique.

2NF: A table is in 2NF when it is already in 1NF and all non-key attributes depend on the full primary k

3NF: A table is in 3NF when it is in 2NF and all non-key attributes depend only on the primary key, not

Example: A StudentCourse table may store student_id, student_name, course_id, course_name, and

This design repeats student and course details. By separating the data into Student, Course, and Enro

A student who enrolls in two courses should not have to repeat the same student information in multip

Normalization improves consistency, reduces update anomalies, and keeps relational data easier to q